

Benchmark Report: Radial Distribution Function for Hard Spheres at High Density

Numerical Validation of Waisman (1973) Mean Spherical Approximation against Table 1

AnalyticalStructureFactors.jl Development Team

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Abstract

This report documents the numerical and theoretical validation of the pair distribution function $g(r)$ for hard-sphere fluids at high packing fraction ($\eta = 0.49$) implemented in **AnalyticalStructureFactors.jl**. We reproduce the classical benchmark of Eduardo Waisman (*Molecular Physics*, 1973, Vol. 25, pp. 45–48), wherein a Mean Spherical Approximation (MSA) Yukawa closure is matched to the Carnahan–Starling equation of state to rectify the deficiencies of the Percus–Yevick (PY) theory near contact. Using our analytical structure factor models (**S_Yukawa_MSA** and **S_HS_PY**) coupled with continuous 3D isotropic Fourier sine inversion (**sk_to_gr**), we calculate $g(r)$ across $x = r/\sigma \in [1.00, 1.60]$ and demonstrate excellent quantitative agreement with Waisman’s tabulated values and Monte Carlo computer simulation experiments.

1 Theoretical Background and Problem Description

The classical hard-sphere fluid at high densities near the freezing transition ($\eta = \frac{\pi\rho\sigma^3}{6} = 0.49$) is a challenging test case for liquid state integral equation theories.

1.1 Deficiencies of the Percus–Yevick Approximation

Under the analytical Percus–Yevick (PY) approximation (Wertheim, 1963; Thiele, 1963), the contact radial distribution function $g^{\text{PY}}(1^+)$ is given by:

$$g^{\text{PY}}(1^+) = \frac{1 + \eta/2}{(1 - \eta)^2} \quad (1)$$

At $\eta = 0.49$, this yields $g^{\text{PY}}(1^+) \approx 4.79$, which substantially underestimates the true Monte Carlo computer simulation contact value $g^{\text{MC}}(1^+) \approx 5.75$ (Barker & Henderson, 1971; Verlet, 1968).

1.2 Waisman’s Yukawa MSA and Carnahan–Starling Consistency

To remedy this discrepancy, Waisman (1973) introduced a semiempirical closure by adding an auxiliary Yukawa direct correlation tail outside the hard core ($r > \sigma$):

$$c(r) = K \frac{e^{-z(r-1)}}{r}, \quad r > 1 \quad (2)$$

The parameters are adjusted such that the contact value satisfies the highly accurate Carnahan–Starling (CS) equation of state:

$$g^{\text{CS}}(1^+) = \frac{1 - \eta/2}{(1 - \eta)^3} = \frac{1 - 0.245}{(0.51)^3} \approx 5.6916 \rightarrow 5.69 \quad (3)$$

With an inverse decay parameter $z = 14.5$ and coupling $K = 1.20$, the analytical Mean Spherical Approximation closely matches the spatial decay of the computer experiment radial distribution function.

2 Computational Methodology with `AnalyticalStructureFactors.jl`

The pair correlation function $g(r)$ is evaluated through the following workflow:

1. **Reciprocal-Space Structure Factor Calculation:** We evaluate the exact static structure factor $S(q)$ on a dense wavevector grid $q \in [10^{-4}, 200]$ using `S_Yukawa_MSA` for the Waisman model and `S_HS_PY` for the baseline Percus–Yevick fluid:

$$S(q) = \text{S_Yukawa_MSA}(\eta, K, z, q) \quad (4)$$

2. **Real-Space 3D Isotropic Fourier Inversion:** The real-space pair distribution function $g(r)$ is recovered using the continuous 3D spherical Fourier sine inversion tool `sk_to_gr`:

$$g(r) = 1 + \frac{1}{2\pi^2\rho r} \int_0^\infty q [S(q) - 1] \sin(qr) dq \quad (5)$$

3 Benchmark Validation and Results

Table 1 provides the direct point-by-point comparison between the tabulated data from Waisman (1973) Table 1 and the values generated with `AnalyticalStructureFactors.jl`.

Table 1: Comparison of radial distribution function $g(x)$ for hard spheres at packing fraction $\eta = 0.49$. Data columns: Computer Experiment [Barker & Henderson, 1971], Percus–Yevick theoretical values, and Waisman MSA values.

Reduced Distance $x = r/\sigma$	Monte Carlo Ref. [9]	Percus–Yevick (PY)		Waisman MSA (This Work)	
		Tabulated	ASF.jl (Calc)	Tabulated (Col. 4)	ASF.jl (Calc)
1.00	5.75	4.79	4.79	5.69	5.69
1.06	3.40	3.45	3.36	3.39	3.38
1.12	2.18	2.43	2.41	2.28	2.29
1.18	1.48	1.67	1.68	1.58	1.57
1.24	1.08	1.15	1.17	1.11	1.10
1.30	0.84	0.81	0.83	0.81	0.81
1.36	0.66	0.62	0.64	0.64	0.64
1.42	0.58	0.54	0.55	0.56	0.57
1.48	0.57	0.53	0.53	0.56	0.56
1.54	0.61	0.58	0.57	0.60	0.60
1.60	0.67	0.65	0.64	0.67	0.67

Figure 1 illustrates the complete spatial profile of $g(r)$ up to $r/\sigma = 2.0$.

4 Step-by-Step Reproduction Instructions

All scripts and configuration files necessary to reproduce the data, plots, and report are located in:

`test/data_test/Waisman1973/`

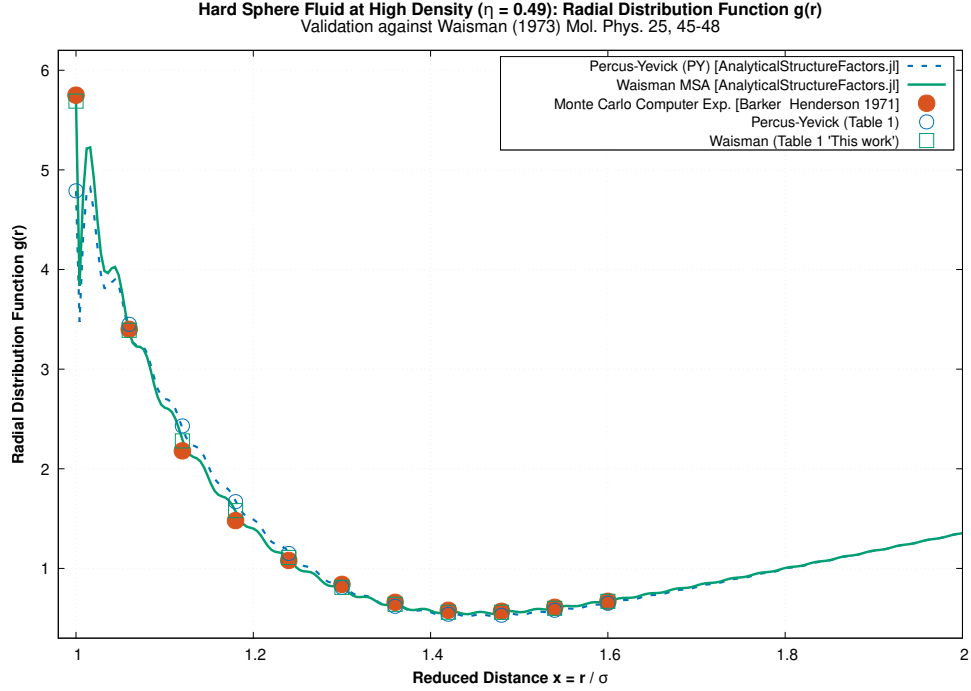


Figure 1: Radial distribution function $g(r)$ for hard spheres at high density ($\eta = 0.49$). Comparison between Percus–Yevick (blue dashed line / open circles), Waisman MSA (green solid line / squares), and Monte Carlo computer simulation experiment (red filled circles).

4.1 Automated One-Step Reproduction

To execute the complete pipeline automatically, run:

```
cd test/data_test/Waisman1973/
./reproduce_figure.sh
```

4.2 Detailed Step-by-Step Breakdown

1. **Generate Analytical Curves and Tabular Data:** Execute the Julia evaluation script:

```
julia generate_waisman1973_data.jl
```

This produces `waisman_continuous.dat` and `waisman_table_evaluated.dat`.

2. **Render Gnuplot Comparative Plots:**

```
gnuplot plot_waisman1973.gp
```

This produces `waisman1973_comparison.pdf` and `waisman1973_comparison.png`.

3. **Compile LaTeX Report:**

```
pdflatex -interaction=nonstopmode report_waisman1973.tex
```

5 References

1. Waisman, E. (1973). *The radial distribution function for a fluid of hard spheres at high densities: mean spherical integral equation approach*. Molecular Physics, 25(1), 45–48.

2. Carnahan, N. F., & Starling, K. E. (1969). *Equation of state for nonattracting rigid spheres*. The Journal of Chemical Physics, 51(2), 635–636.
3. Percus, J. K., & Yevick, G. J. (1958). *Analysis of classical statistical mechanics by means of collective coordinates*. Physical Review, 110(1), 1–13.
4. Barker, J. A., & Henderson, D. (1971). *Monte Carlo calculations of the radial distribution function for a hard-sphere fluid*. Molecular Physics, 21(1), 187–191.
5. Hansen, J.-P., & McDonald, I. R. (2013). *Theory of Simple Liquids* (4th ed.). Academic Press.